Reference Schematics For RK3528

RK3528_BOX_REF_SCH_V10

Main Functions Introduction

1) Power: DiscretePower:BUCK+LDO or PMIC RK805-6
2) RAM: DDR3 4x16bit
   DDR4 2x16Bit
   DDR4 4x16Bit
   LPDDR3 1x32bit
   LPDDR4/LPDDR4X 1x32bit
3) ROM: eMMC5.1 or FSPI Flash
4) Support: Micro SD Card3.0
5) Support: 1 x USB3.0 OTG + 1 x USB2.0 HOST
6) Support: 1 x HDMI2.0 TX
7) Support: 1 x AV OUT
8) Support: SDIO WiFi5 + UART/PCM BT
   PCIE WIFI5 + UART/PCM BT
9) Support: 1 x Ethernet(Embed PHY) + 1x Ethernet(RGMII or RMII)
10) Support: Optical S/PDIF TX
11) Support: IR Receiver
12) Support: Audio-MicArray 2xI2S-DMIC or 4xPDM-DMIC
13) Support: RECOVER/RESET/SARADC_BOOT_KEY,HW_ID
14) Support: 1 x PCIE2.0 Slot
15) Support: Debug UART

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Generate Bill of Materials

Header:

Item\tPart\tDescription\tPCB Footprint\tReference\tQuantity\tOption

Combined property string:

{Item}\t{Value}\t{Description}\t{PCB Footprint}\t{Reference}\t{Quantity}\t{Option}

Description

Note

Option

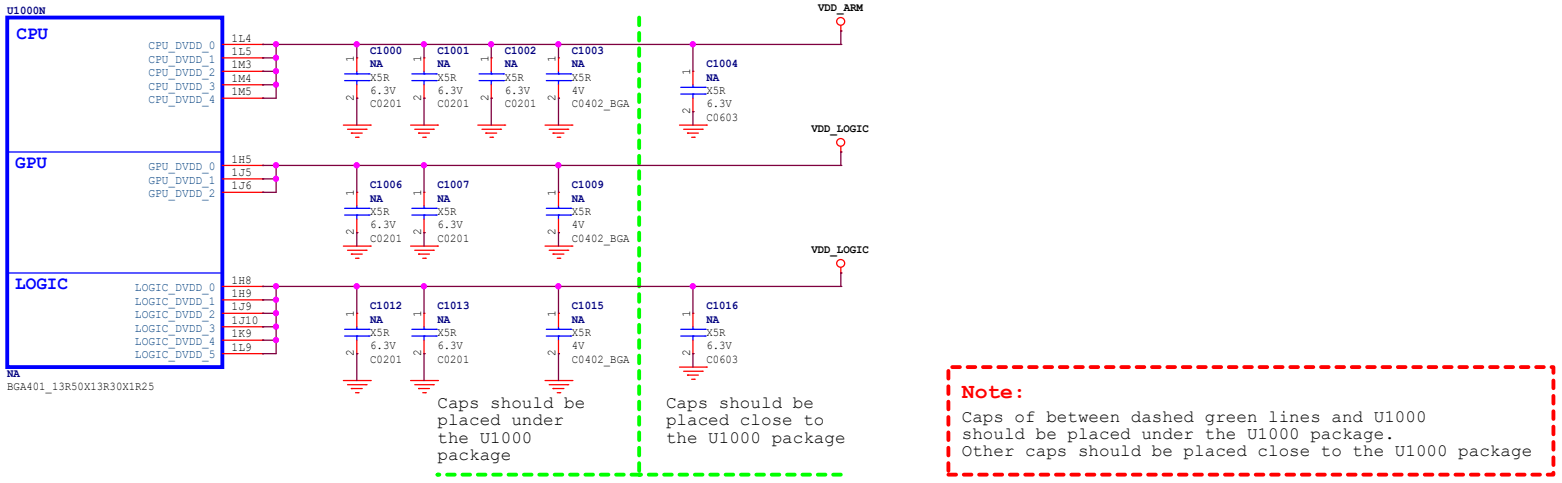
Notes

NOTE 1:
Component parameter description
1. DNP stands for component not mounted temporarily
2. If Value or option is DNP, which means the area is reserved without being mounted

NOTE 2:
Please use our recommended components to avoid too many changes.
For more informations about the second source,please refer to our AVL.

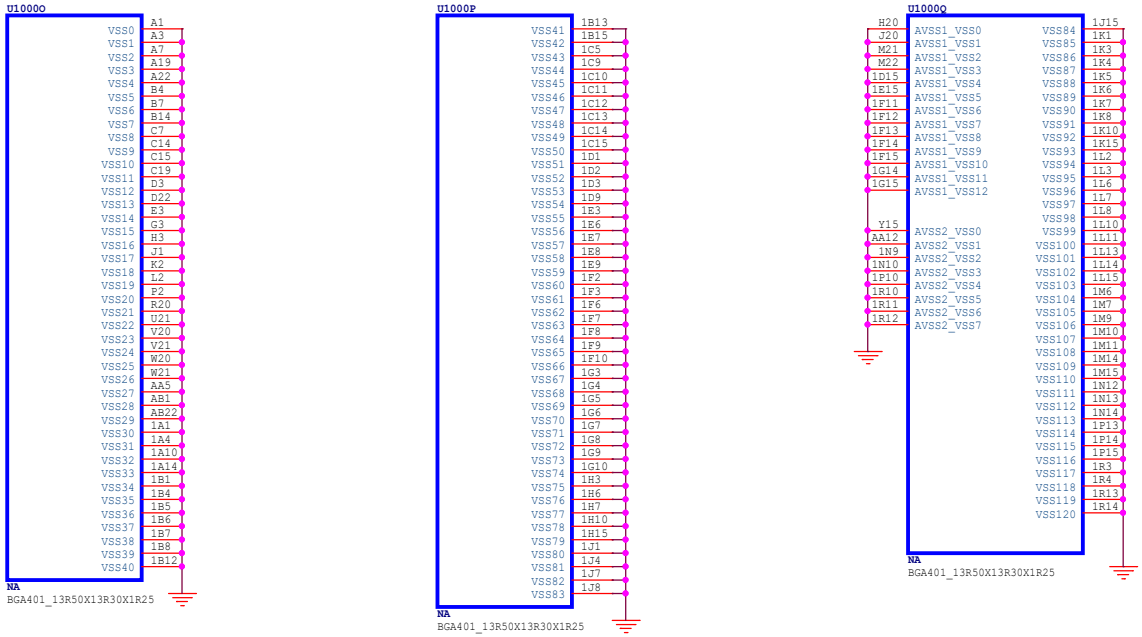
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RK3528_N(POWER)



Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

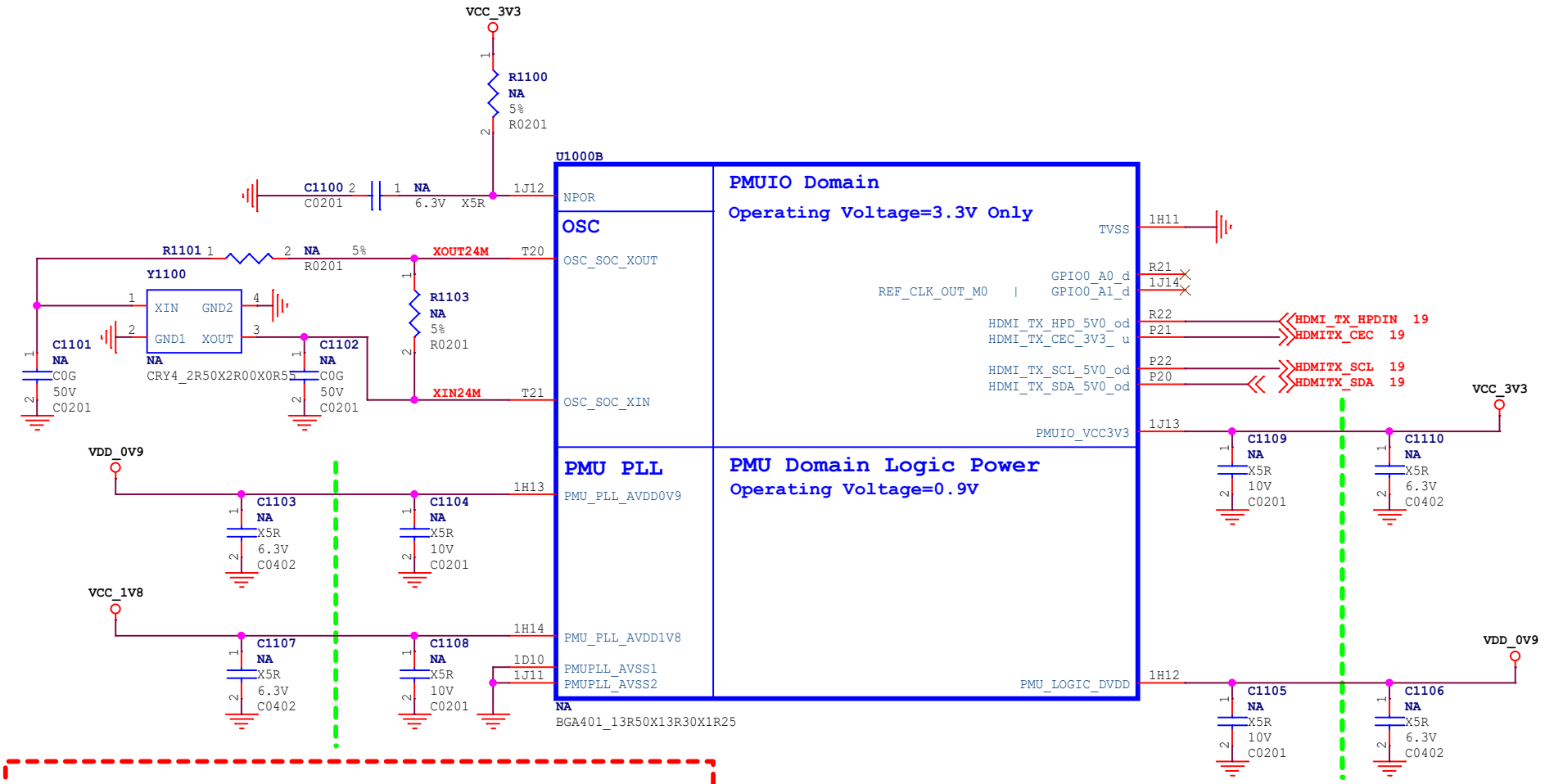
RK3528_O/P/Q(GND)



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RK3528_B(OSC/PLL/PMUIO Domain)

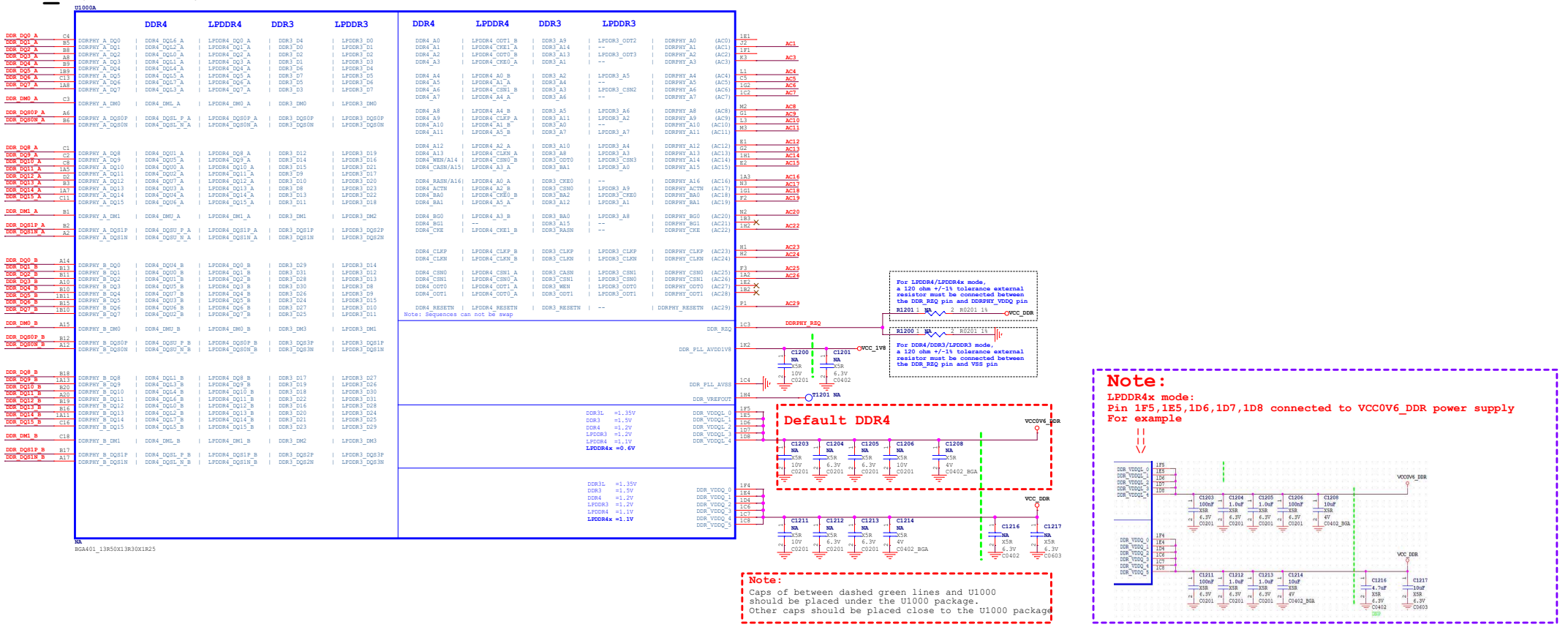


Note:

Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

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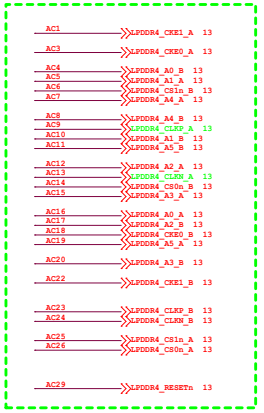
RK3528_A(DDR PHY)



LPDDR4/4x DQ

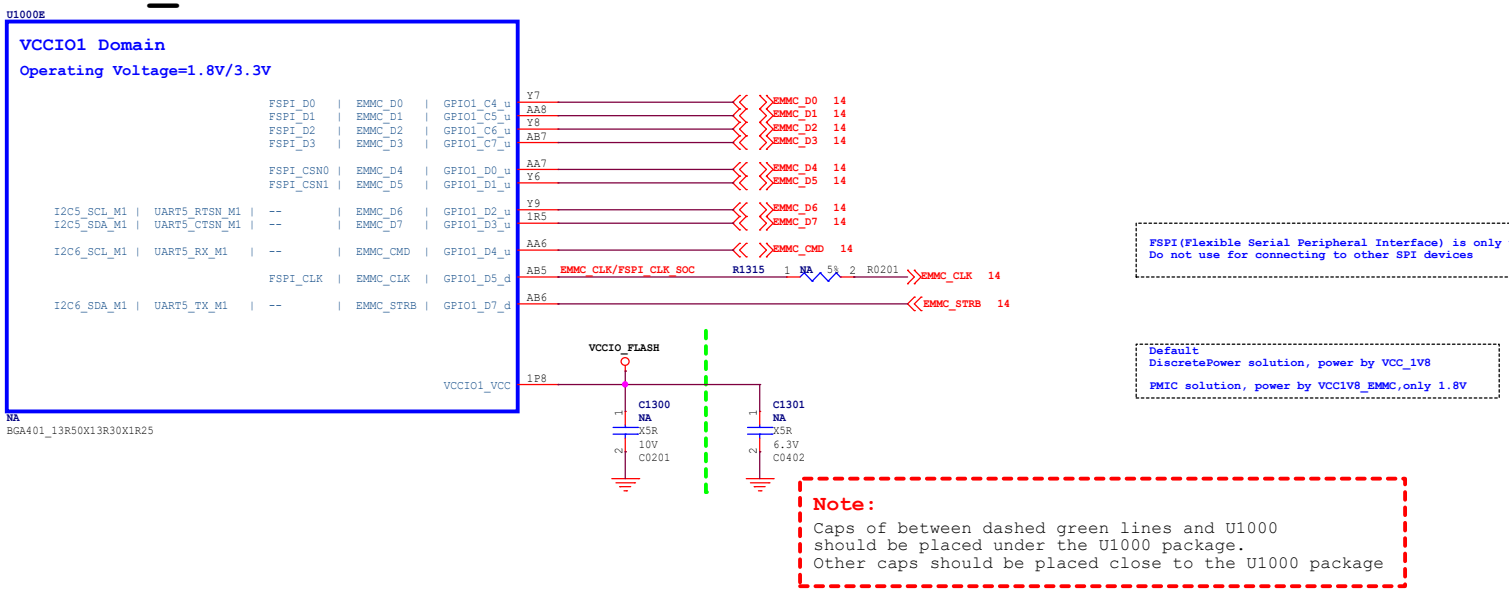


LPDDR4/4x AC



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RK3528_E(VCCIO1 Domain)



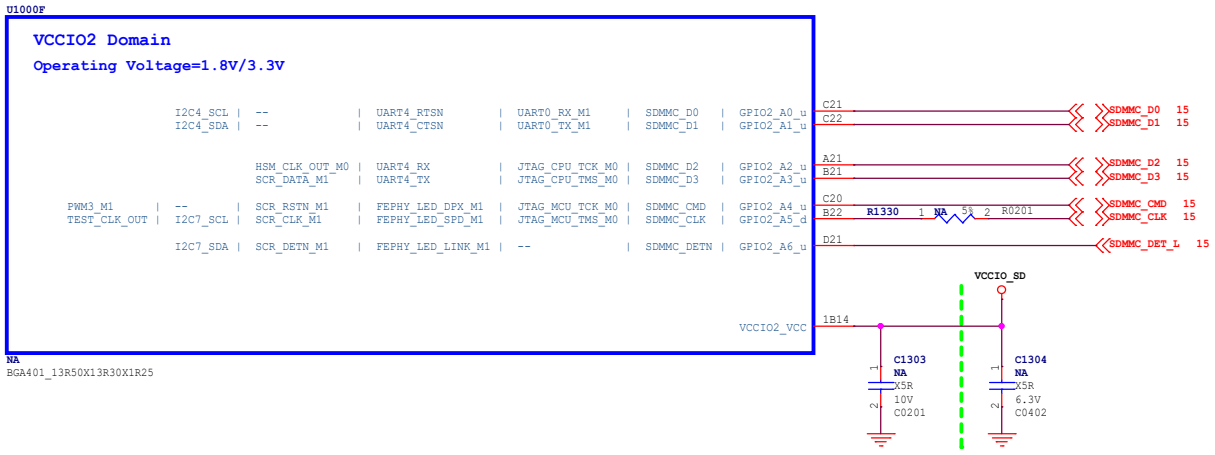
FSPI(Flexible Serial Peripheral Interface) is only used to connect SPI Flash/Memory devices.
Do not use for connecting to other SPI devices

Default
DiscretePower solution, power by VCC_1V8

PMIC solution, power by VCC1V8_EMMC,only 1.8V

Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

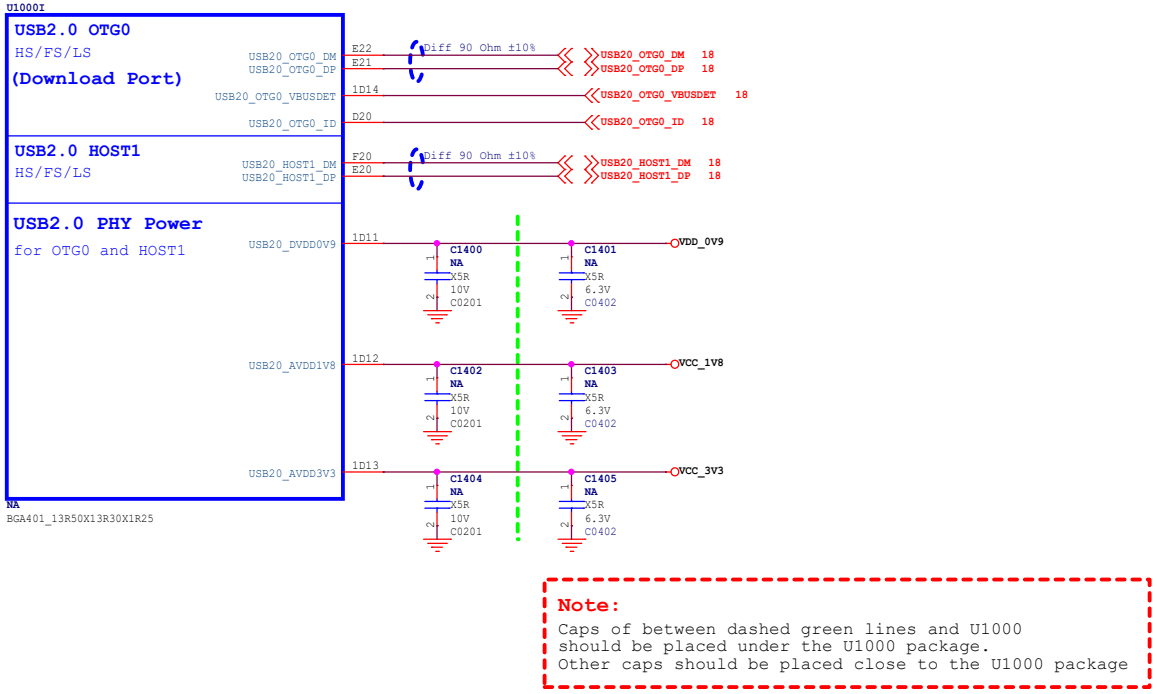
RK3528_F(VCCIO2 Domain)



Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

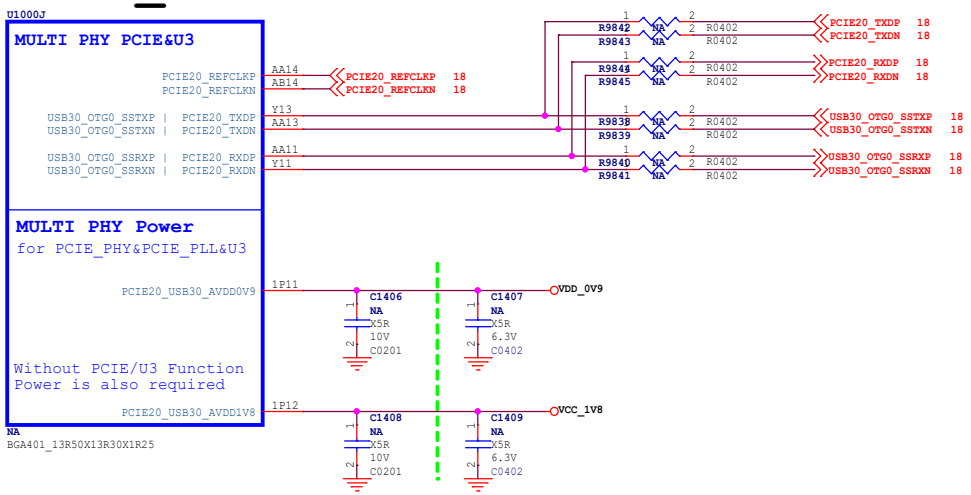
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RK3528_I(USB2.0 OTG/HOST)



Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

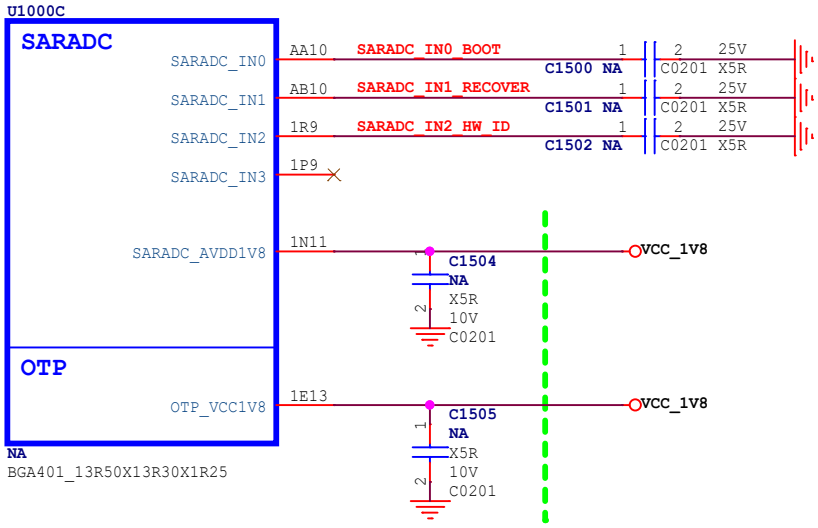
RK3528_J(PCIE2.0/U3 PHY)



Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

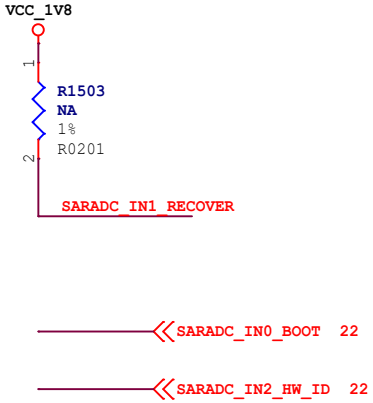
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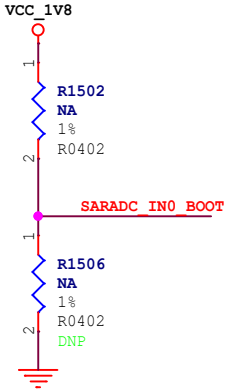
RK3528_C(Saradc/OTP)



Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package



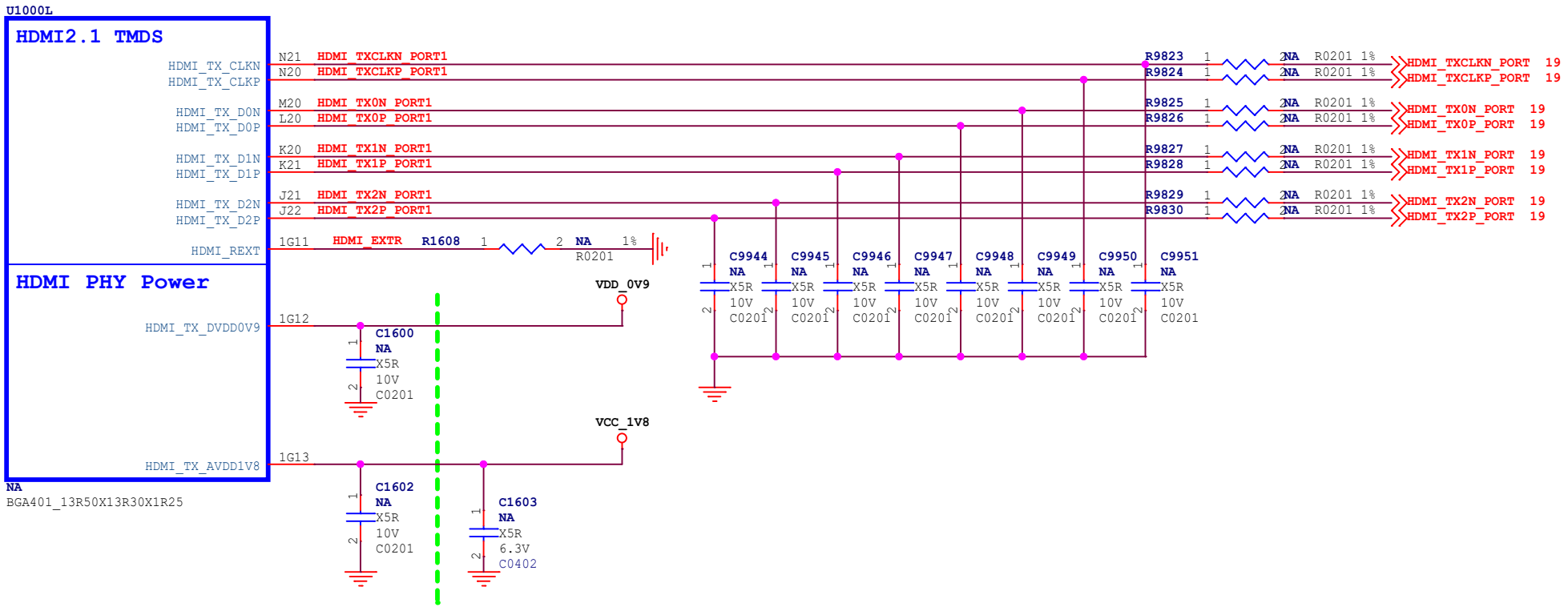


TABLE

Item	Rup	Rdown	ADC	BOOT MODE
LEVEL1	DNP	100K	0	USB (Maskrom mode)
LEVEL2	100K	12K	114	
LEVEL3	100K	27K	228	FSPI--USB
LEVEL4	100K	51K	342	
LEVEL5	100K	82K	456	
LEVEL6	100K	120K	570	EMMC--USB
LEVEL7	100K	200K	683	EMMC--SD Card--USB
LEVEL8	100K	330K	796	SD Card--USB
LEVEL9	100K	820K	910	
LEVEL10	100K	DNP	1023	FSPI--EMMC--SD Card--USB

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RK3528_L(HDMI PHY)

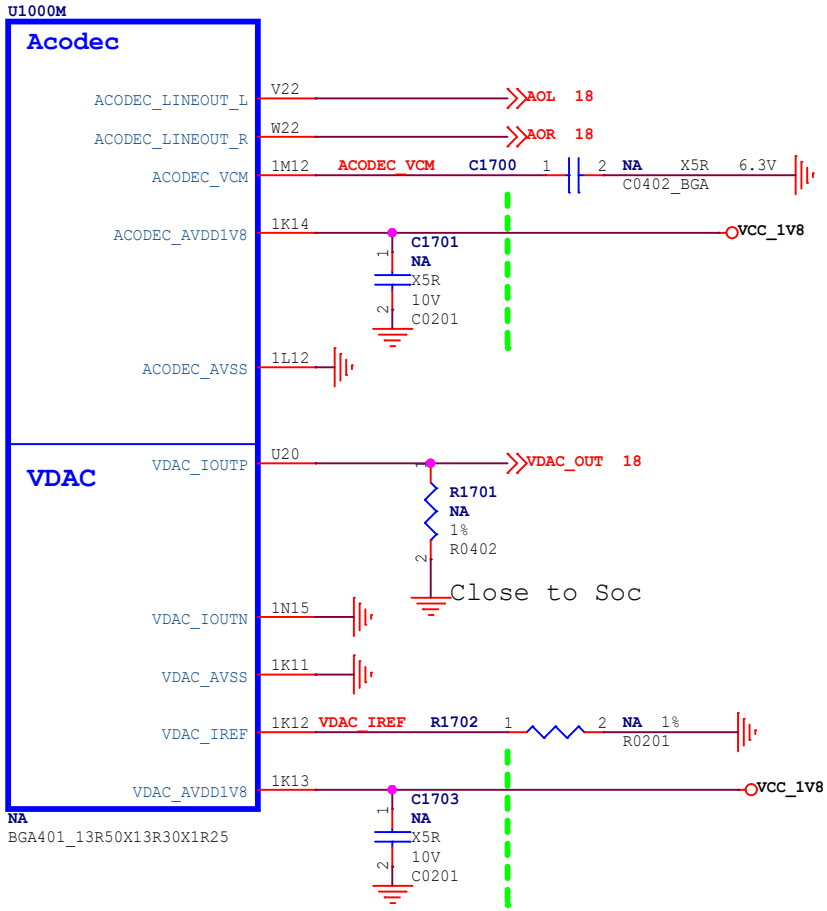


Note:

Caps of between dashed green lines and U1000
should be placed under the U1000 package.
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RK3528_M(Acodec/VDAC)

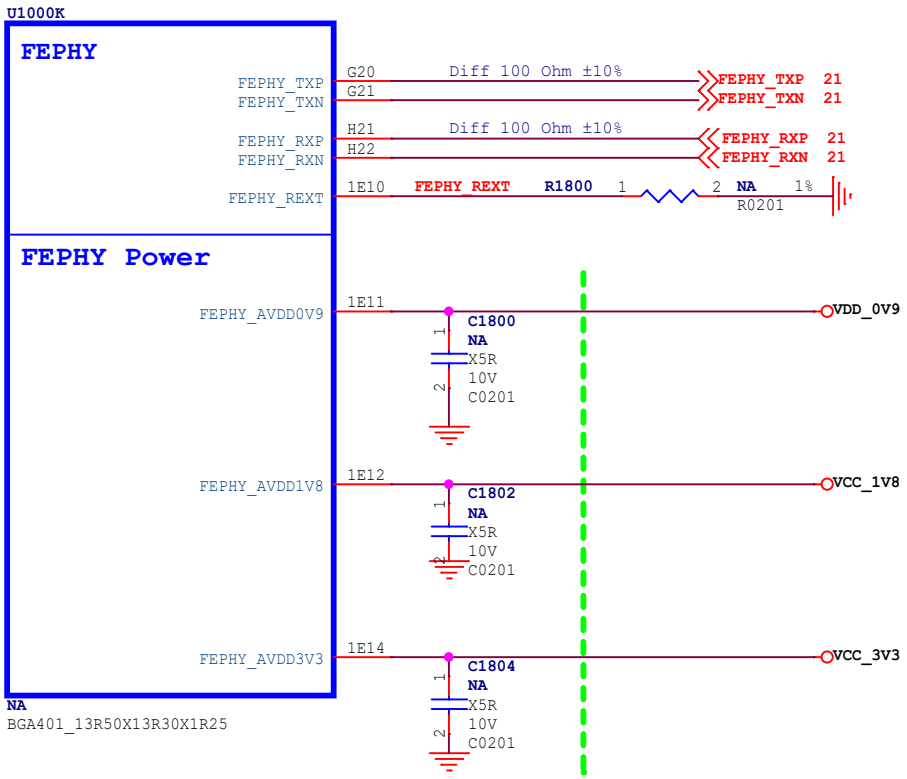


Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package.
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RK3528_K(Embed FEPHY)

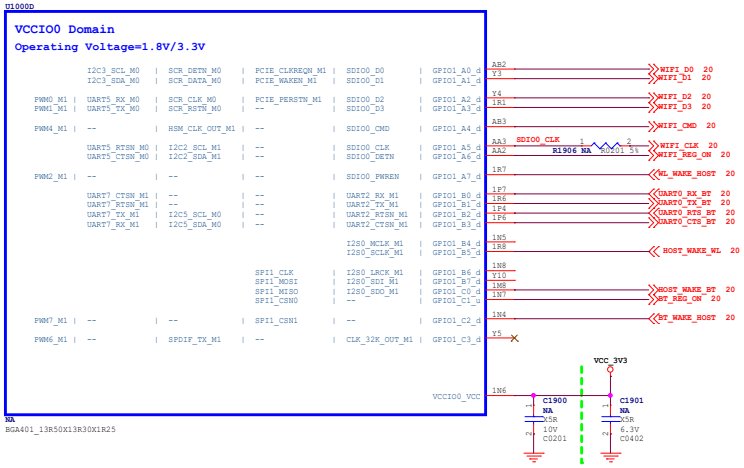


Note:

Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

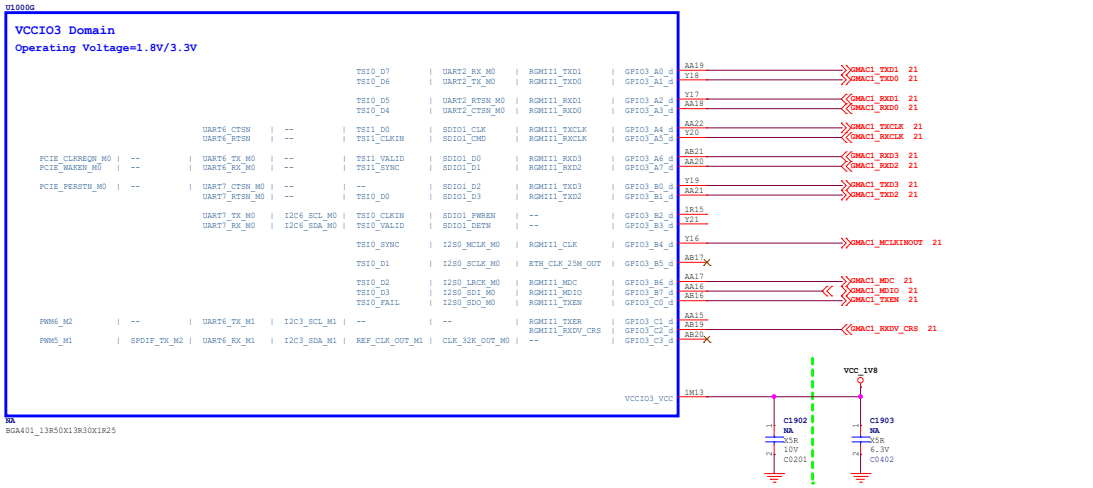
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RK3528_D(VCCIO0 Domain)



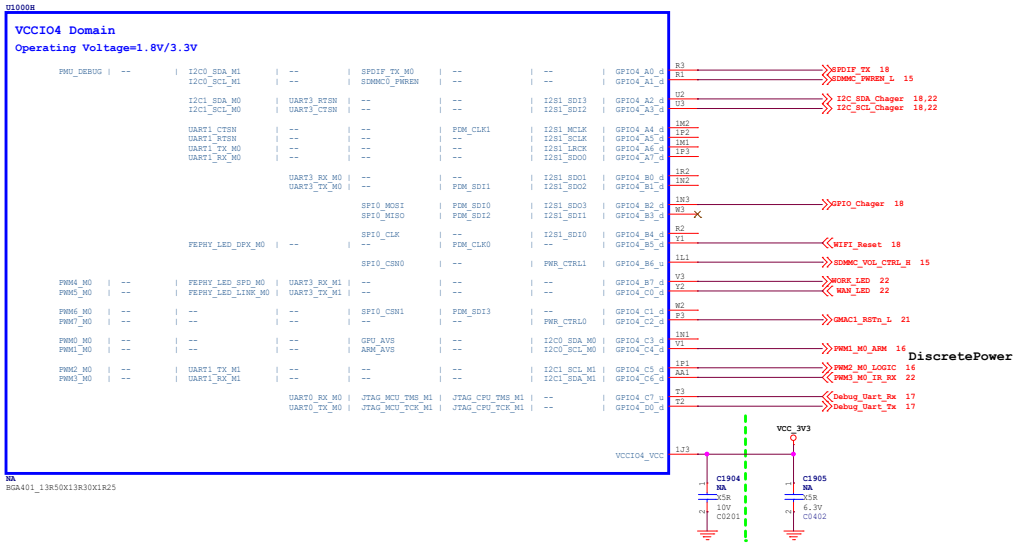
Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3528_G(VCCIO3 Domain)



Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

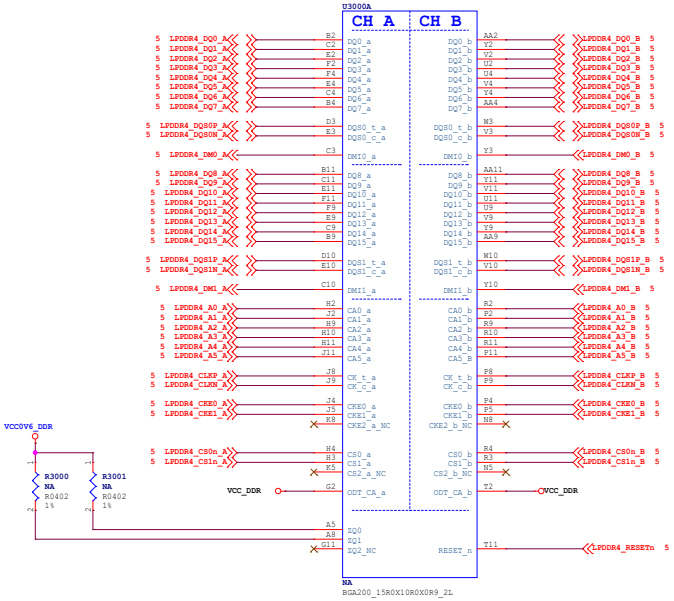
RK3528_H(VCCIO4 Domain)

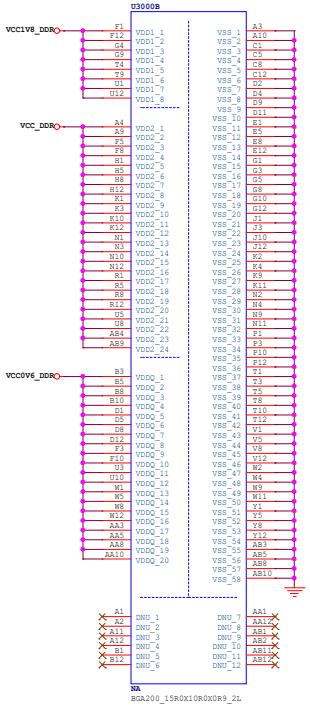


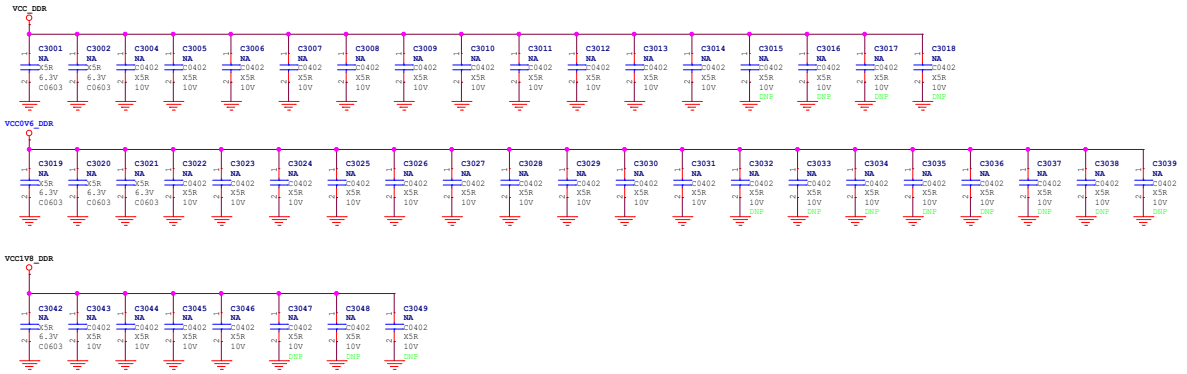
Note:
Caps of between dashed green lines and U1000
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Other caps should be placed close to the U1000 package

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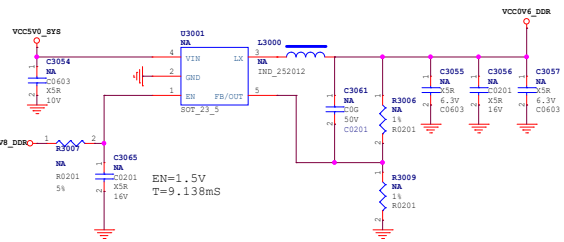
LPDDR4X 200Ball



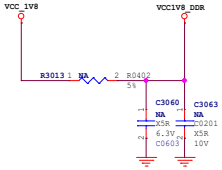




VCC0V6_DDR



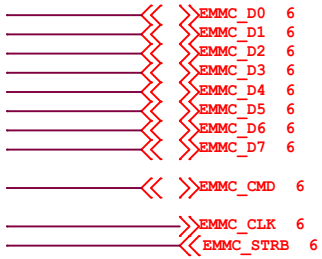
VCC1V8_DDR

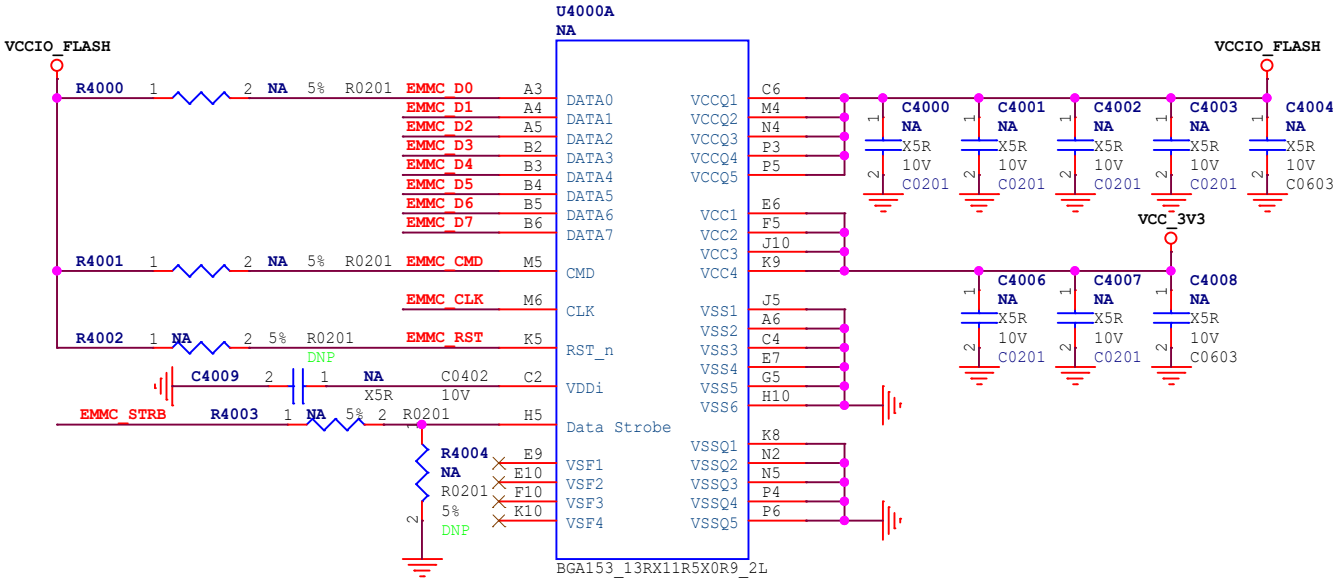


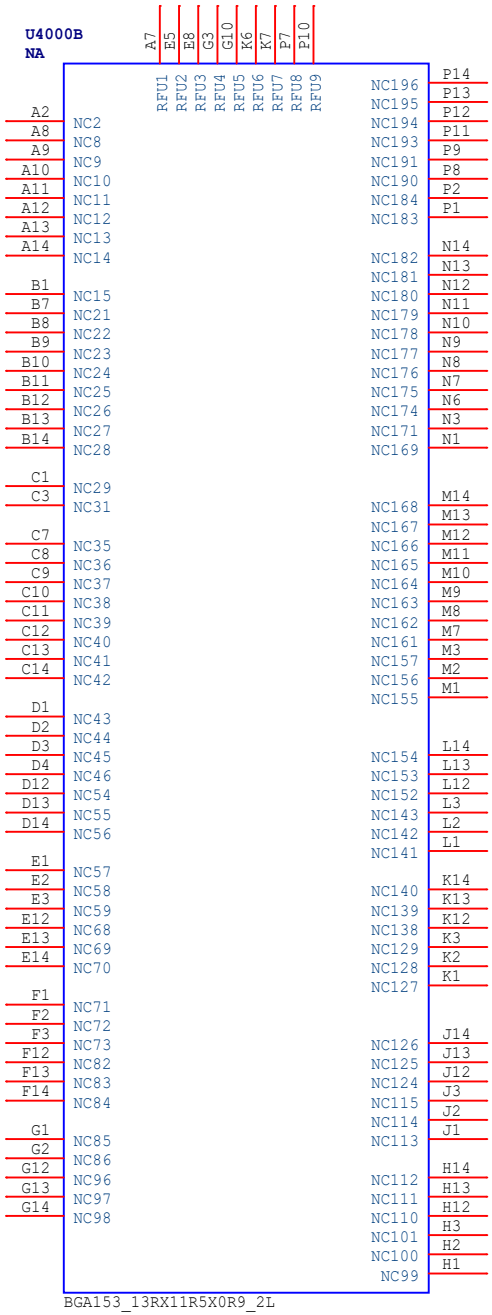
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EMMC

FSPI

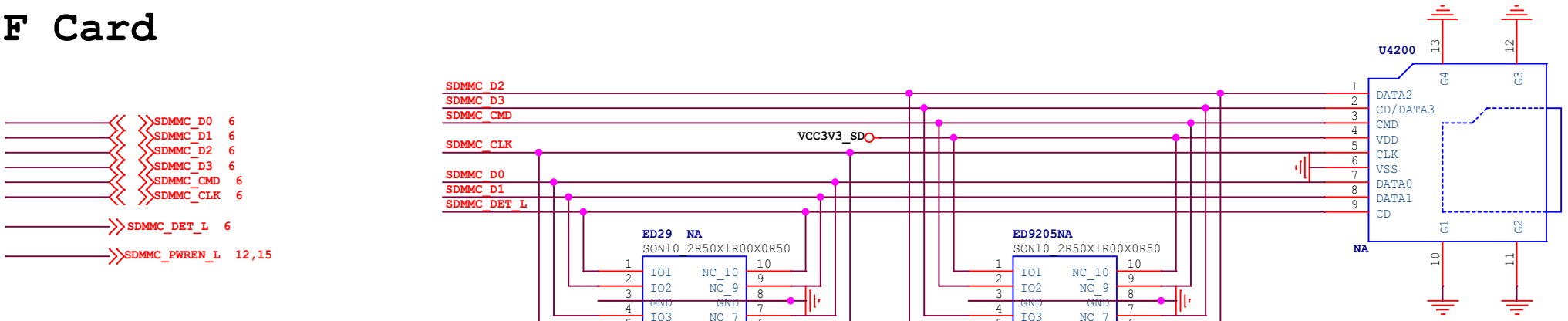




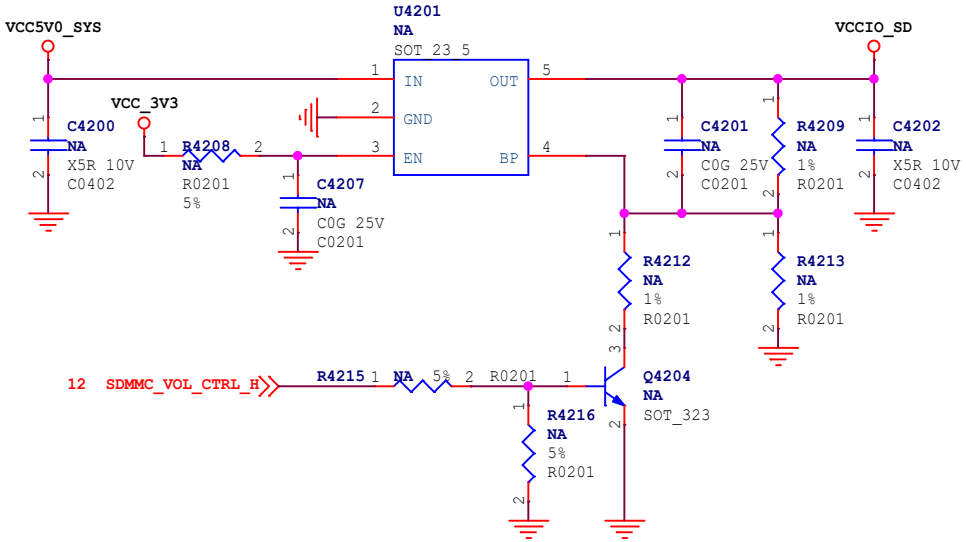


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TF Card

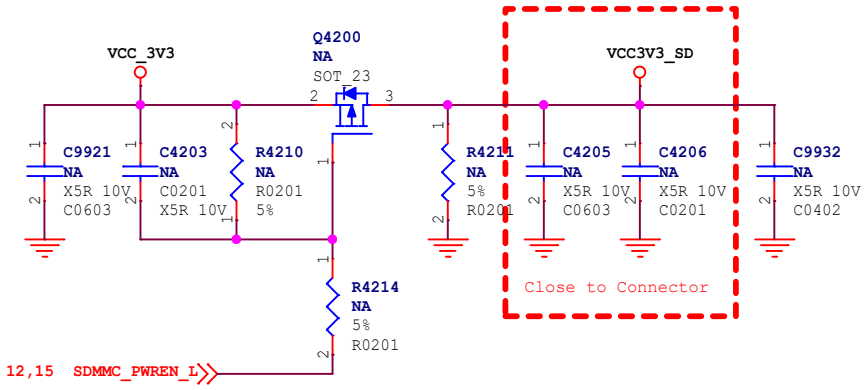


VCCIO_SD



SDIO2.0 SDMMC0_VOL_CTRL=H VCCIO_SD=3.34V(Default)
SDIO3.0 SDMMC0_VOL_CTRL=L VCCIO_SD=1.816V

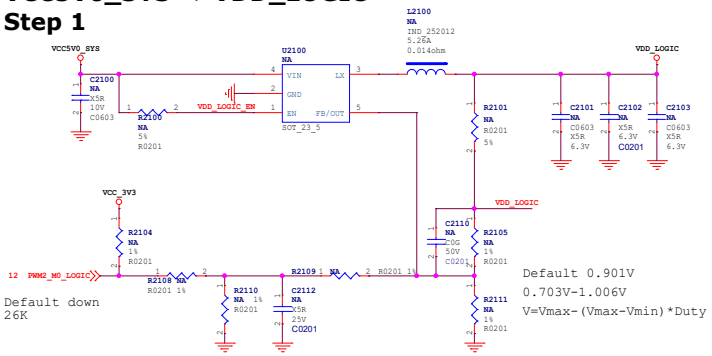
VCC3V3_SD



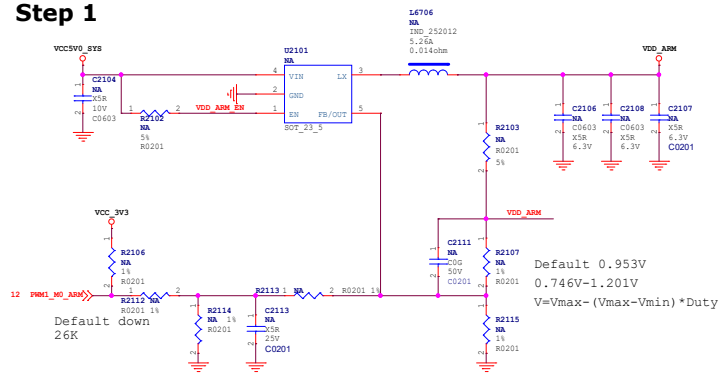
SDMMC_PWREN=L VCC_SD=3.3V(Default)
SDMMC_PWREN=H VCC_SD=0V

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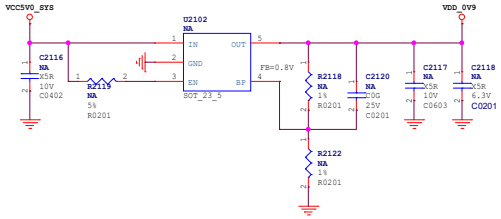
VCC5V0_SYS-->VDD_LOGIC
Step 1



VCC5V0_SYS-->VDD_ARM
Step 1

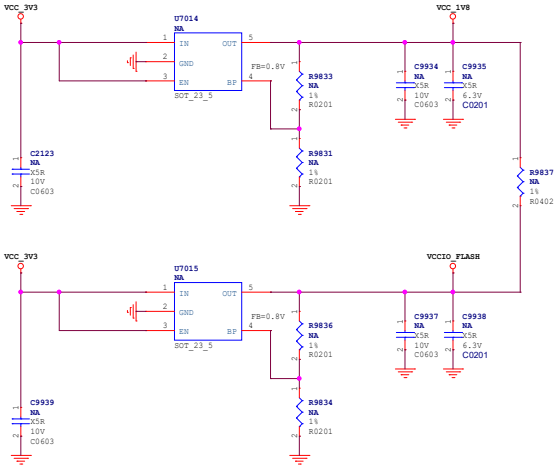


VCC5V0_SYS-->VDD_0V9
Step 1

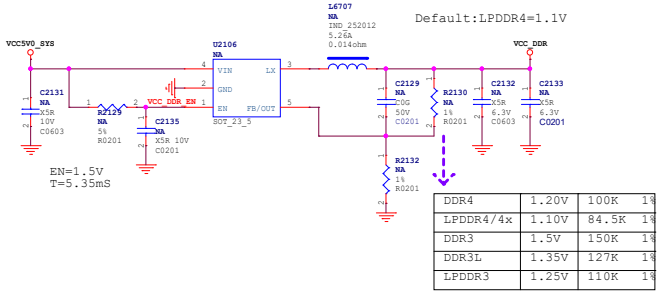


Note:
When LAYOUT, a star must be used,
Branch to each load

VCC_3V3->VCC_1V8(LDO)
Step 2

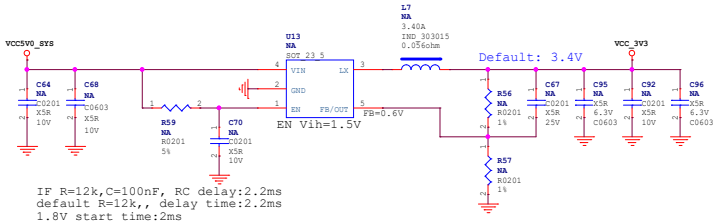


VCC5V0_SYS-->VCC_DDR
Step 2



DDR4	1.20V	100K 1%
LPDDR4/4x	1.10V	84.5K 1%
DDR3	1.5V	150K 1%
DDR3L	1.35V	127K 1%
LPDDR3	1.25V	110K 1%

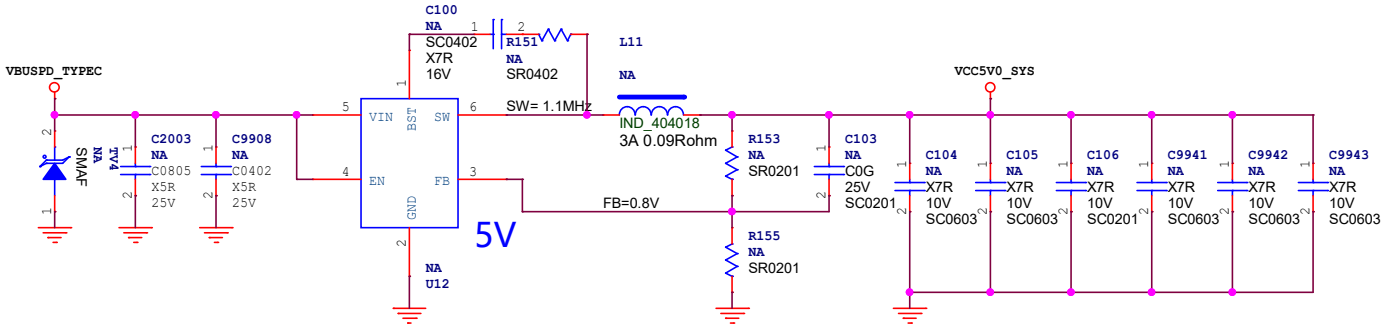
VCC5V0_SYS-->VCC_3V3
Step 2

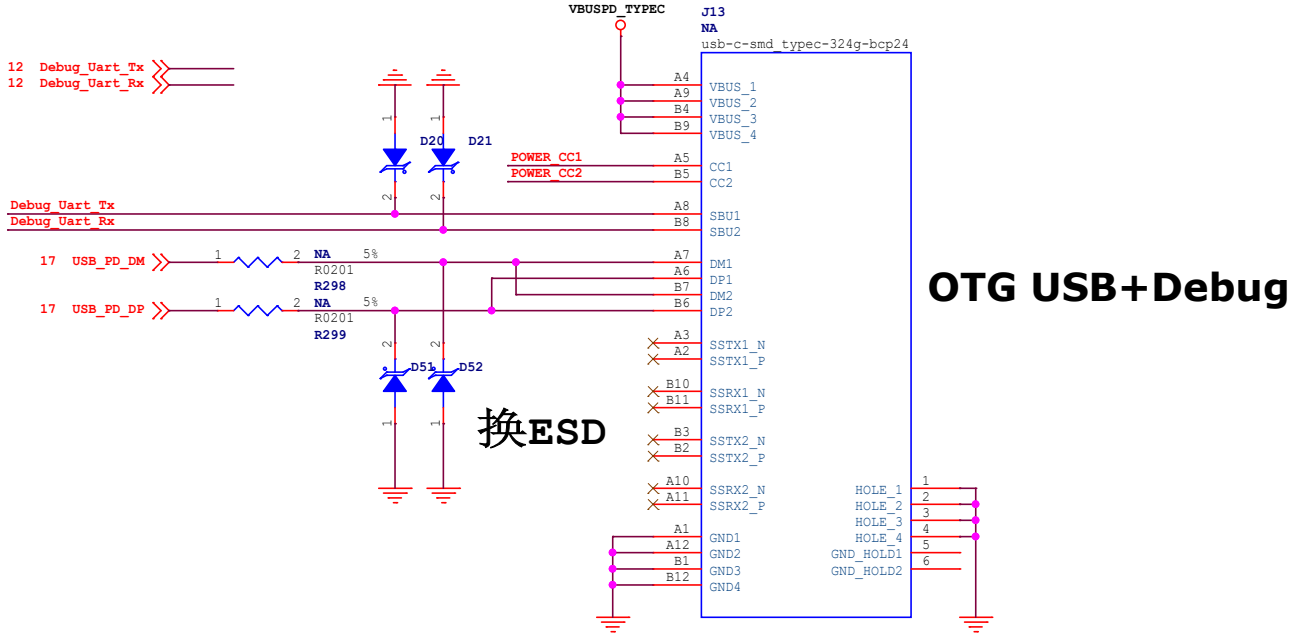


IF R=12k,C=100nF, RC delay:2.2ms
default R=12k,, delay time:2.2ms
1.8V start time:2ms
total delay time=4.2ms

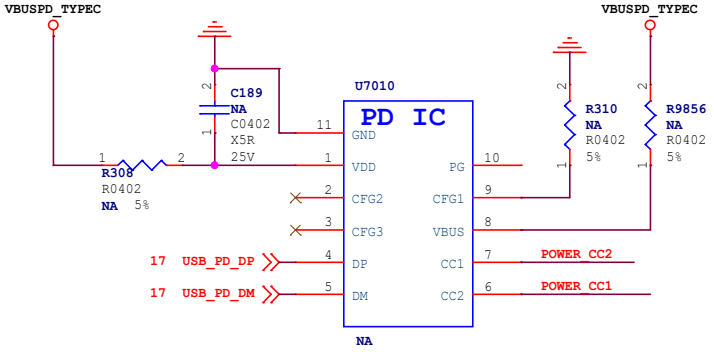
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5V_SYS

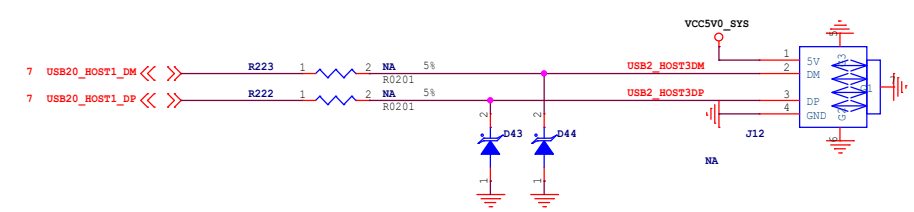


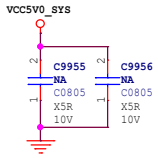


OTG USB+Debug

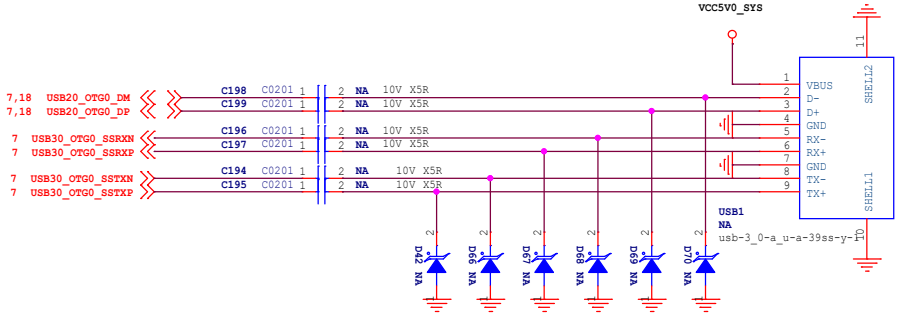


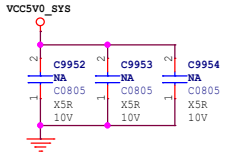
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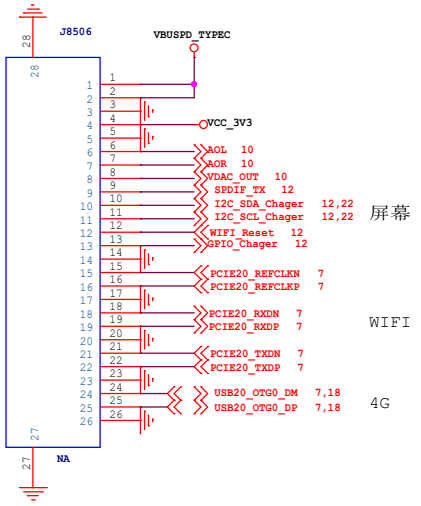


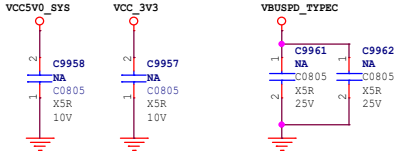
USB2.0 HOST1

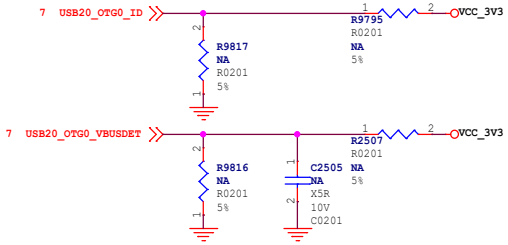




USB3.0 HOST1



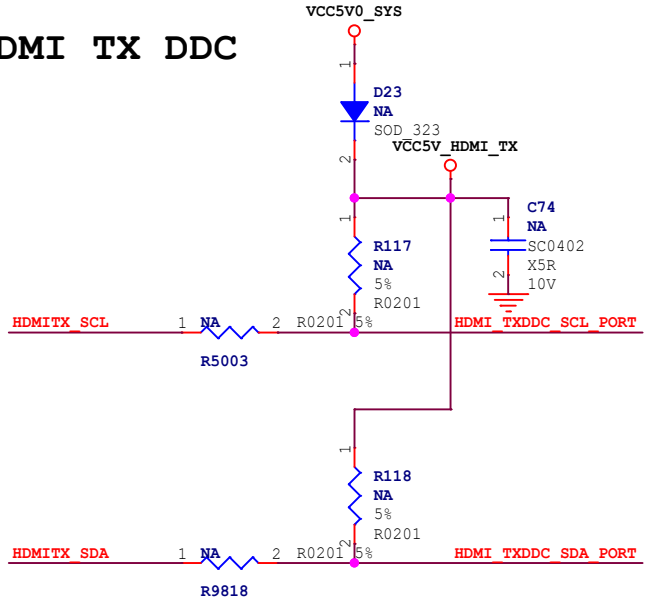


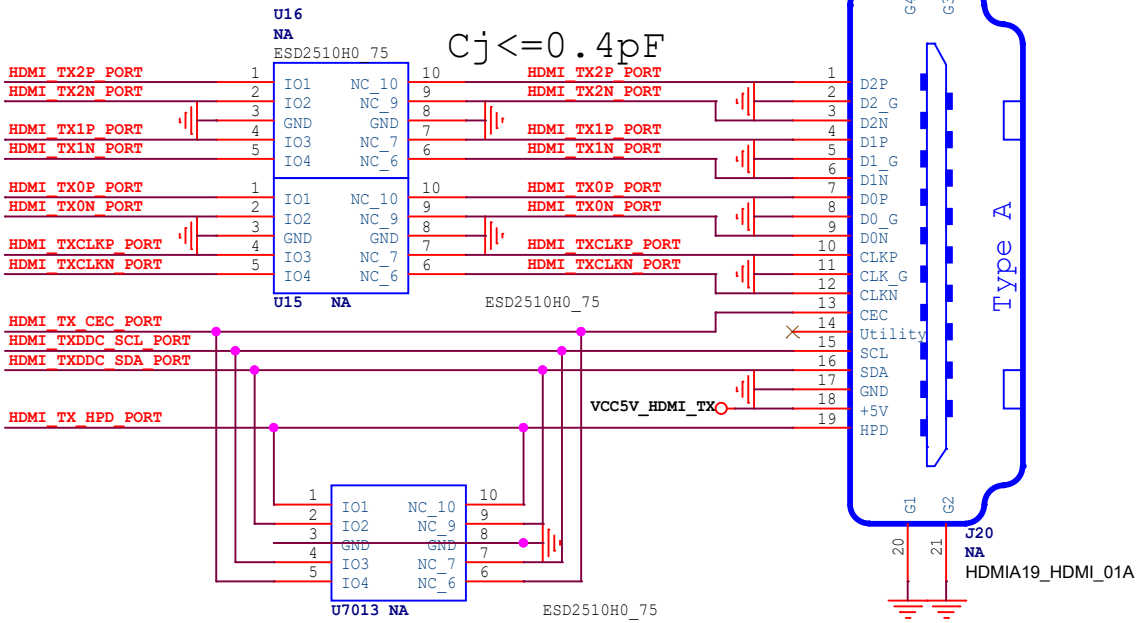


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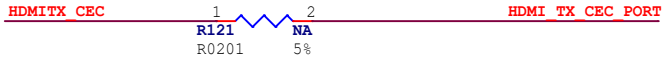
HDMI_TX2P_PORT
HDMI_TX2N_PORT
HDMI_TX1P_PORT
HDMI_TX1N_PORT
HDMI_TX0P_PORT
HDMI_TX0N_PORT
HDMI_TXCLKP_PORT
HDMI_TXCLKN_PORT
HDMITX_SCL
HDMITX_SDA
HDMITX_CEC
HDMI_TX_HPDIN

HDMI TX DDC

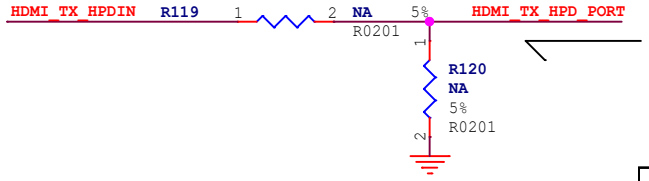




HDMI TX CEC



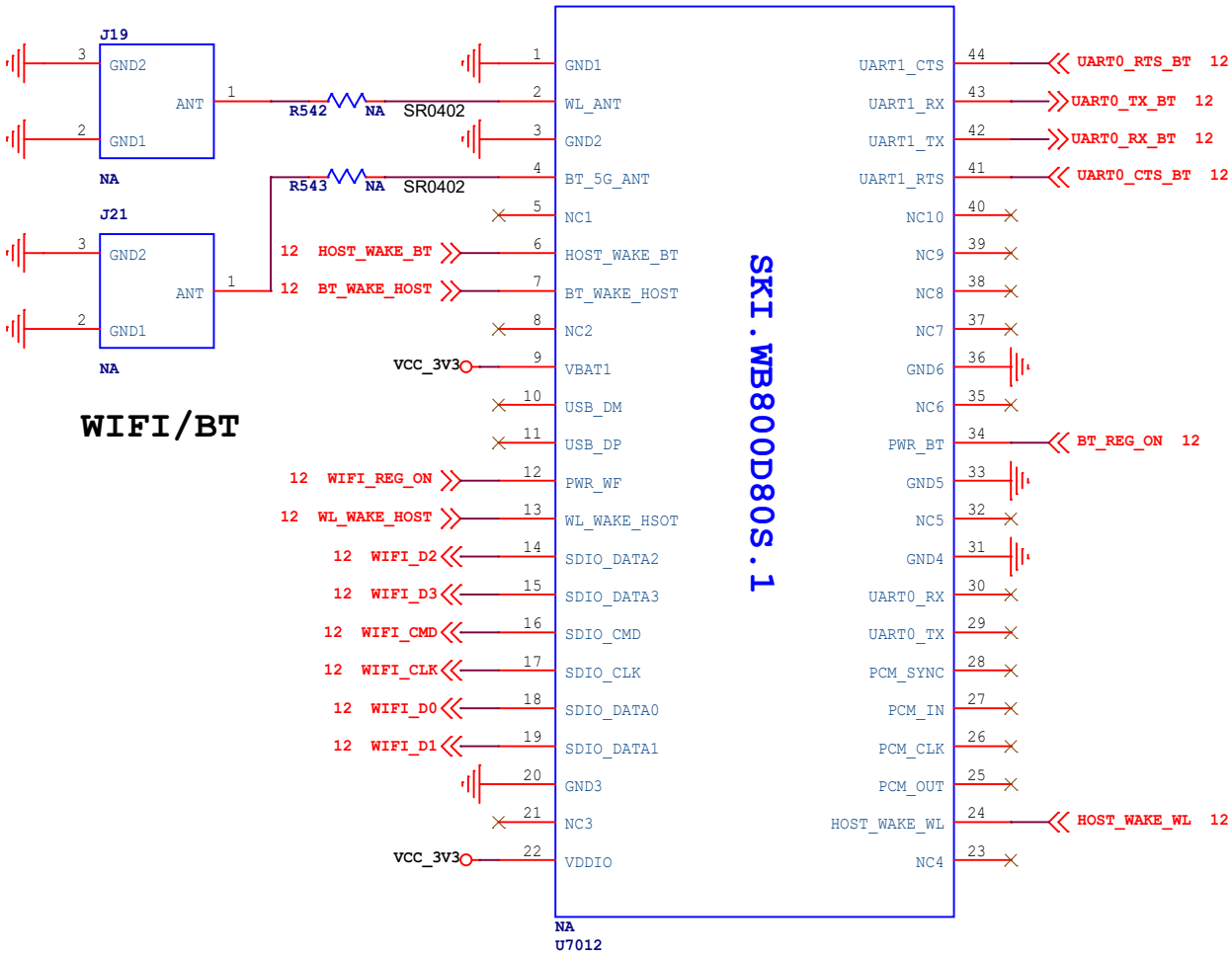
HDMI TX HPD

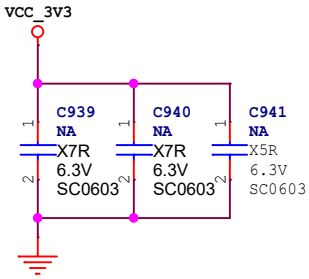


Rockchip Confidential

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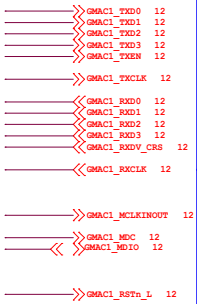
WIFI

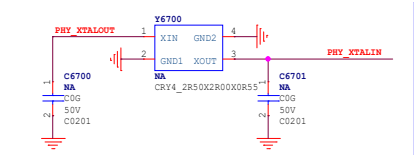


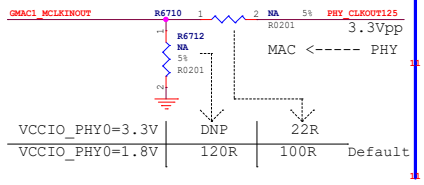


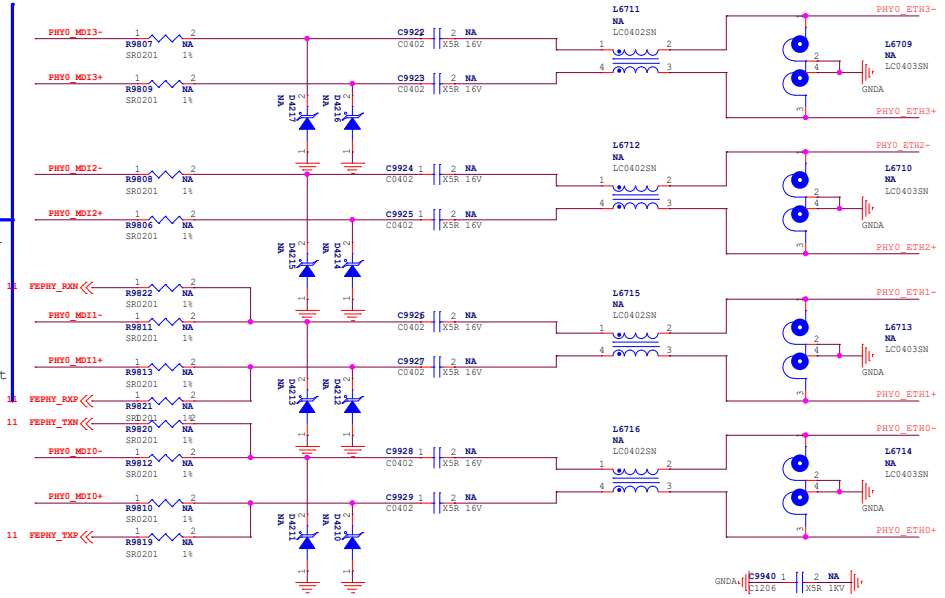
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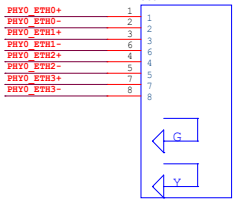
RGMII



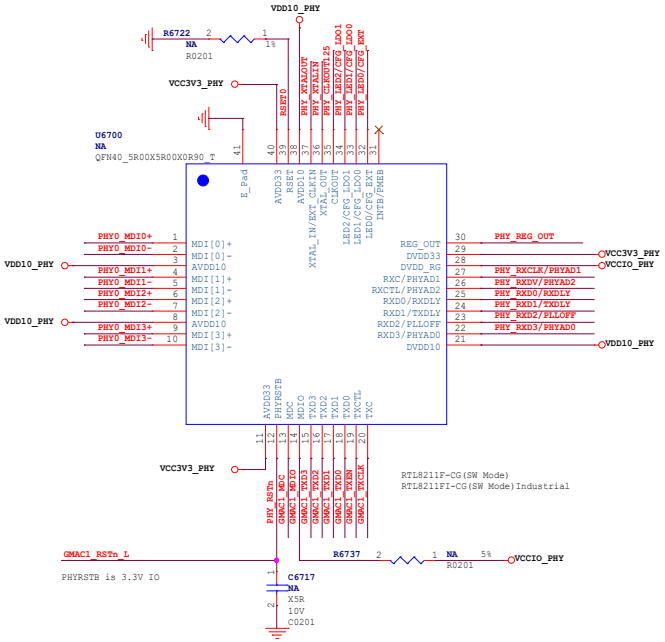


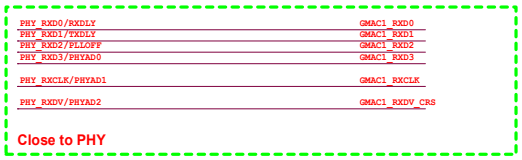


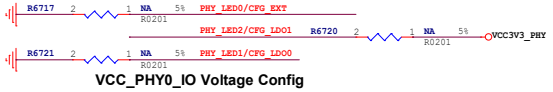


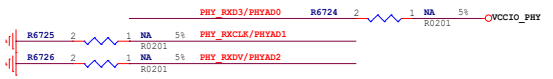


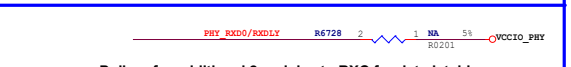
默认不带灯





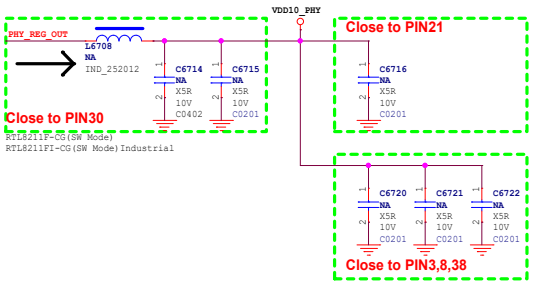




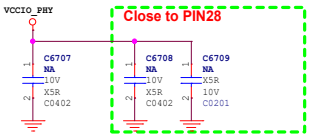


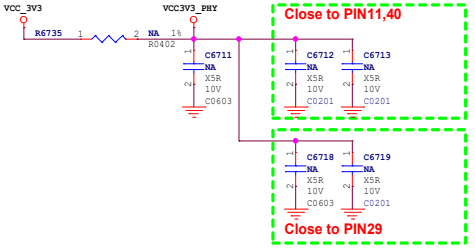






RGMII Power Source	CFG_EXT	CFG_LDO[1:0]	
External 3.3V	1'b1	2'b00	
External 1.8V	1'b1	2'b10	
External 1.8V(default)	1'b0	2'b10	



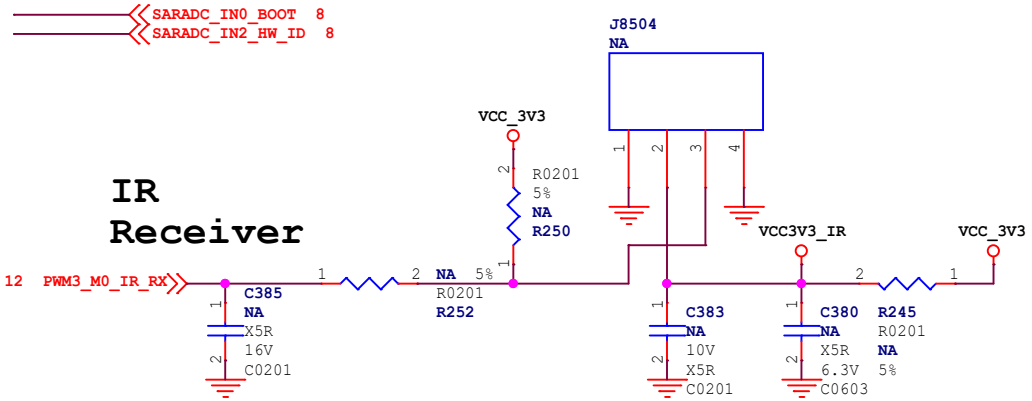


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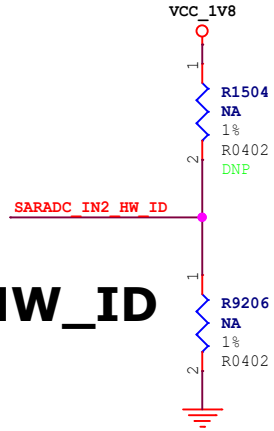
KEY

SARADC_IN0_BOOT 8
SARADC_IN2_HW_ID 8

IR Receiver



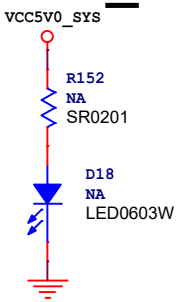
HW_ID



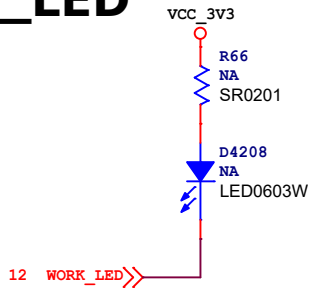
V1.0 **TABLE**

SARADC_IN2	Rup	Rdown	ADC
HW_ID0	DNP	100K	0
HW_ID1	100K	12K	114
HW_ID2	100K	27K	228
HW_ID3	100K	51K	342
HW_ID4	100K	82K	456
HW_ID5	100K	120K	570
HW_ID6	100K	200K	683
HW_ID7	100K	330K	796
HW_ID8	100K	820K	910
HW_ID9	100K	DNP	1023

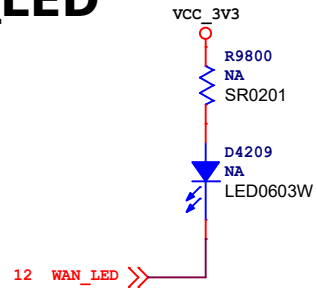
POWER_LED



WORK_LED

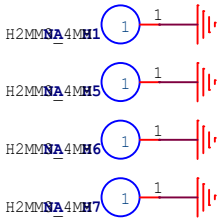


WAN_LED





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MARK4
NA



MARK2
NA



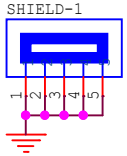
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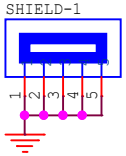
MARK3
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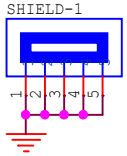
Shield6
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SHIELD-1



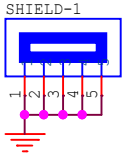
Shield14
NA
SHIELD-1



Shield7
NA
SHIELD-1



Shield13
NA
SHIELD-1



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